

1. Administrative & Study Identification

Full Study Title: A 28-week, randomized, double-blind, active-controlled, multicenter study to evaluate the efficacy of subcutaneously administered secukinumab compared to ustekinumab in adult patients with psoriatic arthritis and failure of TNF α -inhibitor treatment **Short Title/Acronym:** AgAIN Study **Protocol Number:** CAIN457FDE04 / 2019-004246-15 **NCT Number:** NCT04632927 **Posting ID:** 685088376386540482 **Sponsor/Company Name:** Novartis Pharmaceuticals **Investigational Product:** Secukinumab (Cosentyx) and Ustekinumab (Trade Name: Stelara | ATC Code: L04AC05 | [EMA Product Information](#)) **Phase of Development:** PHASE3 **Trial Status:** COMPLETED **Medical Monitor:** Novartis Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the safety and efficacy of secukinumab against ustekinumab by evaluating the proportion of patients achieving a Health Assessment Questionnaire - Disability Index (HAQ-DI) response at Week 28. **Secondary Objectives:** To evaluate improvements in Patient's Assessment of Pain on VAS, tender joint count (TJC 68), swollen joint count (SJC 66), PASI 75 and PASI 90 response rates, Leeds Dactylitis Index (LDI), and disease-specific quality of life (PsaQoL) at Week 28.

2.2 Background & Rationale To compare the efficacy of targeting IL-17A (secukinumab) versus IL-12/23 (ustekinumab) in adults with active psoriatic arthritis who have experienced an inadequate response or intolerance to previous TNF α -inhibitor treatment. This head-to-head trial generates critical comparative evidence to support biologic sequencing decisions in psoriatic disease management.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Ustekinumab) **Blinding:** Triple-blind (Participant, Investigator, Outcomes Assessor) **Randomization:** 1:1 Parallel Assignment

3.2 Planned Enrollment & Duration Target \$N\$: 119 patients **Number of Sites:** 28 centers across Germany **Estimated Study Start:** 21-Dec-2020 **Trial Status:** Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Diagnosis of PsA as classified by CASPAR criteria for at least 6 months before randomization. 2. Active PsA at baseline defined as ≥ 3 tender joints out of 68 and ≥ 3 swollen joints out of 66 (dactylitis of a digit counts as one joint each). 3. Inadequate response or intolerance to previous or current treatment with at least one TNF α inhibitor. 4. Inadequate response or intolerance to conventional disease modifying anti-rheumatic drugs (cDMARDs). 5. Diagnosis of active plaque psoriasis, with at least one psoriatic plaque of ≥ 2 cm diameter and/or nail changes consistent with psoriasis and/or documented history of plaque psoriasis. 6. Rheumatoid factor (RF) and anti-cyclic citrullinated peptide (CCP) antibodies negative at screening.

4.2 Exclusion Criteria 1. Pregnant or nursing women. 2. Previous exposure to secukinumab, ustekinumab, or any other biologic drug directly targeting IL-17, IL-17 receptor, IL-12, or IL-23. 3. Patients for whom the use of secukinumab or ustekinumab is contraindicated. 4. Use of any other investigational drug. 5. Previous treatment with any cell-depleting therapies including but not limited to anti-CD20 or investigational agents. 6. Evidence of ongoing infectious or malignant processes. 7. Subjects receiving high potency opioid analgesics. 8. Ongoing use of prohibited psoriasis treatments/medications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Secukinumab or ustekinumab according to randomized assignment arm. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 28 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of conventional disease-modifying anti-rheumatic drugs (cDMARDs). **Prohibited:** Cell-depleting therapies, high-potency opioid analgesics, other investigational drugs, and any other biologic therapies targeting IL-17 or IL-12/23.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, Rheumatoid factor (RF) and CCP antibody screening, Baseline assessments
Baseline	Day 0	Randomization (1:1), First SC Dose, Clinical Questionnaires	Interim Weeks 4 to 24 Efficacy Assessments (TJC 68, SJC 66, PASI, Pain VAS), Safety Labs
End of Study	Week 28	Final HAQ-DI, DLQI, PsAQoL, LDI scoring, AE Reporting, Trial Completion	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of Patients With Health Assessment Questionnaire - Disability Index (HAQ-DI) Response at Week 28. **Measurement Method:** The disability assessment component of the HAQ evaluated a subject's level of functional ability through 20 questions grouped into 8 categories: dressing, rising, eating, walking, hygiene, reach, grip, and usual activities. Each item began with the prompt "Over the past week, are you able to..." and is scored on a 4-point scale: 0 (no difficulty), 1 (some difficulty), 2 (much difficulty), and 3 (unable to do). The overall score was calculated by averaging the scores across all answered domains, resulting in a total score ranging from 0 (no disability) to 3 (severe disability). A HAQ-DI response was defined as an improvement of ≥ 0.35 points from baseline at Week 28. Missing values were imputed as non-responders.

7.2 Secondary & Exploratory Endpoints 1. Number of Participants Achieving Psoriasis Area and Severity Index (PASI) 90 Response at Week 28 (and PASI 75). 2. Change From Baseline in Patient's Assessment of Pain on VAS at Week 28. 3. Change from baseline in Tender Joint Count (TJC 68) at Week 28. 4. Change from baseline in Swollen Joint Count (SJC 66) at Week 28. 5. Change from baseline in Leeds Dactylitis Index (LDI) at Week 28.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection of treatment-emergent adverse events (TEAEs), upper respiratory tract infections (URTIs), and vital signs via clinical visits. **Serious Adverse Events (SAEs):** Evaluated throughout the 28-week study period. **Data Monitoring Committee (DMC):** Internal safety oversight provided by Novartis Pharmaceuticals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary outcomes. **Sample Size Justification:** $N = 119$ to provide adequate statistical power for the comparative superiority assessment between the two active biologic arms. **Handling of Missing Data:** Missing binary endpoint values (e.g., HAQ-DI response, PASI response) are imputed as non-responders (Non-Responder Imputation, NRI).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Routine onsite and remote monitoring per sponsor guidelines to assure triple-blind integrity. **Audit Trail:** Assured via Electronic Case Report Forms (eCRF) and centralized data validation processes.

11. Study Identifiers & Governance

Primary ID: CAIN457FDE04 **Registry Number:** NCT04632927 **Posting ID:** 685088376386540482 **Sponsor/Lead Organization:** Novartis **Study Title:** AgAIN Study **Official Regulatory Title:** A 28-week, randomized, double-blind, active-controlled, multicenter study to evaluate the efficacy of subcutaneously administered secukinumab compared to ustekinumab in adult patients with psoriatic arthritis and failure of TNF α - inhibitor treatment

12. Clinical Framework (Psoriatic Arthritis Specific)

Primary Condition: Psoriatic Arthritis (Disease Areas: Psoriatic Arthritis) **Preferred Name:** Arthritis, Psoriatic (Semantic Type: Disease or Syndrome) **Preferred UMLS Name:** Psoriatic Arthritis, Refractory **Definition:** Joint inflammation associated with psoriasis. **Ontology Codes:** MeSH Code: D015535 | SNOMED CT: 156370009 **Diagnosis Threshold:** Active PSA (≥ 3 tender and ≥ 3 swollen joints) and CASPAR criteria ≥ 6 months **Medical Methodology:** Targeted biologic therapies (IL-17 vs IL-12/23 inhibitors) **Optimization Target:** HAQ-DI functional response & resolution of joint/skin inflammation

13. Study Procedures & Methodology

Management Pathway: Head-to-head active comparator assessment **Education Protocol (HCP):** Blinded assessment standardization (TJC 68, SJC 66, LDI, PASI scoring, Pain VAS) **Education Protocol (Patient):** Instruction on patient-reported outcome measures (HAQ-DI, PsAQoL, DLQI) **Quality Audit Mechanism:** Triple-blind tracking to ensure neutral objective comparative evaluation

14. Observation & Follow-Up Schedule

Total Duration: 28 weeks **Onsite Visit Cadence:** Baseline, periodic intermediate visits, and final Week 28 visit **Remote Monitoring Frequency:** Standard safety check-ins **Checklist Review Interval:** Data lock at Week 28 primary completion

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					